Minimum first-order detuning residual for qubit π gates at fixed duration

Tingyu Zhang[1, *] and Yan Cen[1, †]

[1] *Department of Physics, Fudan University, Shanghai, China*

Suppressing static detuning errors in a quantum gate requires additional control time, which can increase the loss from dissipation. For a full qubit π gate driven by arbitrary measurable transverse controls with joint amplitude bound Ω, we determine the smallest first-order detuning residual at every duration between the shortest nominal gate and the shortest first-order robust gate. The bound is attained only by a single-axis three-segment pulse family and its overall negative. This family continuously connects a rectangular π pulse to short-CORPSE, and establishes $7\pi/(3\Omega)$ as the minimum time for first-order detuning cancellation over the full two-axis control class. The proof uses a geometric projection onto the midpoint noise direction and avoids classifying control extremals. The result yields a sharp duration-dependent lower bound on the coefficient of the quadratic term in the small-detuning expansion of average gate infidelity. With weak, drive-independent Markovian dissipation, optimizing the duration selects the best compensation level. Near complete cancellation, shortening the gate reduces dissipative loss linearly while the detuning penalty grows quadratically. Partial compensation reduces the leading gate error by up to 4.91% relative to the better of a rectangular pulse and short-CORPSE. As the relative dissipative cost increases past a threshold, the optimal duration jumps to that of the rectangular pulse.

I. INTRODUCTION

A gate that implements its target exactly on resonance can still respond at first order to a static detuning. Composite pulses suppress this response by canceling the error accumulated during different parts of the drive, at the cost of a longer duration [1, 2]. This creates a resource question: how much detuning error can be removed within a given time, and how much time is needed to remove its first-order contribution completely?

Gate fidelity susceptibility and robustness–duration trade-offs have been formulated in general quantum-control settings [3]. Related speed limits concern robust identity cycles under full control with prescribed error subspaces [4]. For single-qubit gates, numerical optimization has estimated the minimum times needed at prescribed robustness orders [5]. Partial detuning compensation is also a practical design variable in Rydberg-gate optimization [6]. A continuous relation between duration and residual error would allow the compensation level to be selected from the available time or from competition with other noise sources.

The required time depends on whether robustness is imposed on one state or on the full gate. For first-order detuning-robust population inversion, Van Damme et al. reported a minimum time of $2\pi/\Omega$ [7]. For a full qubit π gate, the known short-CORPSE sequence achieves complete first-order detuning cancellation in time $7\pi/(3\Omega)$ [2]. It is optimal within a symmetric family of three elementary pulses [8]. Pontryagin analyses have addressed selective and robust rotations under single-axis control [9] and identified short-CORPSE as a time-optimal candidate [10], while numerical optimization gives first-order robust X-gate times near $2.33\pi/\Omega$ [5].

With the full nominal π gate imposed exactly and two transverse controls subject to a joint amplitude bound, the main research questions are:

1. What is the smallest first-order detuning residual at each prescribed duration?
2. Does the known three-segment pulse family attain the global minimum over all measurable two-axis controls?
3. Which controls attain this minimum?

The geometric representation of detuning errors turns these questions into a problem about curves with bounded curvature [11–13]. The planar three-arc return with reversed tangent occurs in the controllability construction of Sigalotti and Chitour [14], and its shortest length $7\pi/3$ is given explicitly by Kim and Cheong [15]. For planar curves at fixed duration, minimizing the residual amounts to finding the reachable position nearest the origin in the Dubins model with terminal heading opposite to the initial heading [16, 17]. Spatial shortest-path and reachability problems admit additional, nonplanar extremals [18–21]. A planar construction therefore establishes an achievable duration, but a lower bound for the full two-axis control class must also account for nonplanar curves.

The main results are:

1. We determine the minimum first-order detuning residual over the entire duration interval from the shortest rectangular π pulse to short-CORPSE.
2. The attaining controls are exactly the single-axis three-segment family and its overall negative. The first zero of the minimum residual gives the shortest first-order robust gate time, $7\pi/(3\Omega)$.
3. The midpoint projection method gives the bound in every dimension $d \geq 2$ without classifying control

* 23300190021@m.fudan.edu.cn
† Corresponding author: cenyan@fudan.edu.cn

extremals. Its equality conditions force the curve into one plane, and the full gate condition fixes the control axis.

4. The residual bound fixes the minimum quadratic detuning coefficient in average gate infidelity. Together with the first-order effect of weak Markovian dissipation [22], it determines the compensation level that minimizes the leading gate error and its switching threshold.

II. CONTROL MODEL AND MINIMUM DETUNING RESIDUAL

The control model specifies both the available drives and the complete unitary target. We then express its first-order response to detuning as the endpoint displacement of an error curve, which allows the minimum residual to be stated geometrically.

We allow two transverse drives and no independent longitudinal control. Changing the drive phase changes only the subsequent transverse Hamiltonian; it does not apply an additional instantaneous gate. Reduced amplitudes, zero-control intervals, and arbitrary phase changes are allowed. The detuning is constant during each gate, and the controls do not depend on this unknown detuning. With $\hbar = 1$ and Pauli matrices of eigenvalues ± 1, the propagator obeys

$$\begin{aligned} \dot{U}_\delta(t) &= -iH_\delta(t)U_\delta(t), \qquad U_\delta(0) = I, \\ H_\delta(t) &= \frac{u_x(t)\sigma_x + u_y(t)\sigma_y + \delta\sigma_z}{2}. \end{aligned} \tag{1}$$

The real controls $u = (u_x, u_y)$ are measurable and satisfy

$$u_x(t)^2 + u_y(t)^2 \le \Omega^2 \quad \text{for almost every } t, \tag{2}$$

where $\Omega > 0$ is the fixed maximum drive amplitude.

Let $R_x(\theta) = \exp(-i\theta\sigma_x/2)$. At zero detuning we require the full nominal gate

$$U_0(T) = \pm R_x(\pi). \tag{3}$$

The sign allows an overall phase. This operator identity fixes the gate's action on every input state, so matching a single state transfer is insufficient.

To express the first-order detuning residual geometrically, set $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ and define

$$\begin{aligned} \mathbf{n}(t) \cdot \boldsymbol{\sigma} &= U_0(t)^\dagger \sigma_z U_0(t), \\ \mathbf{r}(t) &= \int_0^t \mathbf{n}(t')\,\mathrm{d}t'. \end{aligned} \tag{4}$$

The unit vector $\mathbf{n}(t)$ is the detuning direction in the ideal rotating frame, and $\mathbf{r}(t)$ accumulates that direction over time. This is the error-curve representation used in geometric robust control [12, 13]. Differentiating the evolution with respect to the constant detuning gives

$$\partial_\delta U_\delta(T)|_0 = -\frac{i}{2} U_0(T)\,\mathbf{r}(T) \cdot \boldsymbol{\sigma}. \tag{5}$$

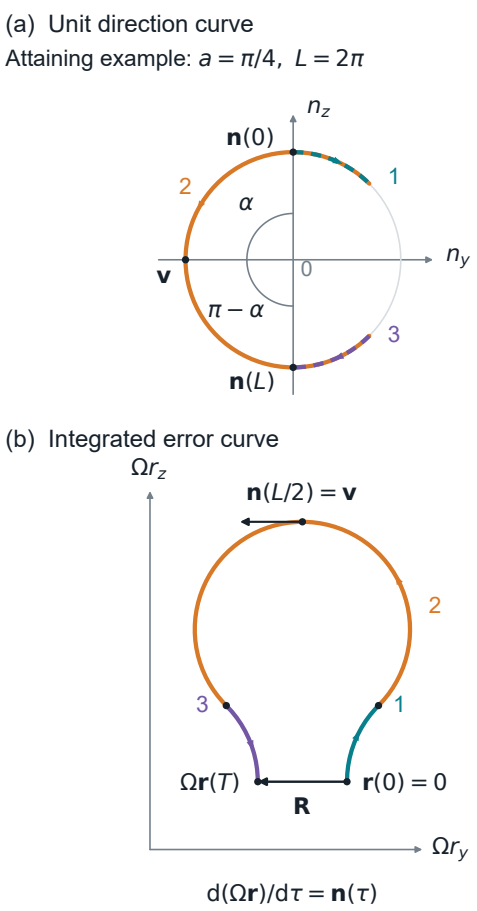


FIG. 1. Direction curve and accumulated first-order detuning error for a representative attaining path, with $a = (L - \pi)/4 = \pi/4$ and $L = 2\pi$. (a) For the same-axis three-segment pulse [10], the unit direction $\mathbf{n}(\tau)$ lies on a great circle in the yz plane. Its endpoints are antipodal, $\mathbf{v} = \mathbf{n}(c)$ at $c = L/2$, and their distances to $\mathbf{v}$ are α and $\pi - \alpha$; here $\alpha = \pi/2$. The dashed first and third segments lie on the same arcs as the solid second segment, traversed in opposite directions; gaps in the dashed traces reveal the overlapping second segment. (b) Integrating this direction gives the dimensionless error curve $\Omega\mathbf{r}$, whose unit tangent is $\mathbf{n}$. Its endpoint displacement is $\mathbf{R} = \Omega\mathbf{r}(T) = -(2\sqrt{2} - 2)\mathbf{e}_y$. Numbers 1–3 and colors identify the same successive pulse segments in both panels; arrows indicate traversal and tangent directions. The midpoint bound applies also to nonplanar curves.

Thus the endpoint displacement $\mathbf{r}(T)$ determines the full first-order response. Canceling that response is equivalent to closing the error curve, $\mathbf{r}(T) = 0$.

Unitary conjugation preserves the Pauli inner product, so $|\mathbf{n}| = 1$ and the error curve has unit speed, $|\dot{\mathbf{r}}| = 1$. Let $\mathbf{a}_0, \mathbf{b}_0, \mathbf{n}$ be the orthonormal vectors obtained by conjugating $\sigma_x, \sigma_y, \sigma_z$, respectively. The commutator in the derivative of (4) yields

$$\dot{\mathbf{n}} = u_x \mathbf{b}_0 - u_y \mathbf{a}_0, \qquad |\dot{\mathbf{n}}| = \sqrt{u_x^2 + u_y^2} \le \Omega \quad \text{a.e.} \tag{6}$$

Writing $\mathbf{e}_z$ for the unit vector along z, the target imposes

$$\mathbf{n}(0) = \mathbf{e}_z, \qquad \mathbf{n}(T) = -\mathbf{e}_z. \tag{7}$$

The shortest path on the unit sphere between these antipodal directions has length π. Since the direction turns at speed no greater than Ω, every admissible gate requires $\Omega T \geq \pi$. For bounded measurable controls, U_0 and $\mathbf{n}$ are Lipschitz, $\mathbf{r} \in W^{2,\infty}$, and the differential identities hold almost everywhere. In particular, the model permits self-intersections of the error curve and discontinuous changes in its curvature at pulse switches.

Define $L = \Omega T$ and rescale time by $\tau = \Omega t$, retaining $\mathbf{n}$ for the rescaled direction. The dimensionless residual is

$$\mathbf{R} = \int_0^L \mathbf{n}(\tau)\, \mathrm{d}\tau = \Omega \mathbf{r}(T). \tag{8}$$

For inversion of a z-axis eigenstate, first-order robustness requires $\mathbf{R} \parallel \mathbf{e}_z$, since this residual changes the state's phase without changing its Bloch vector. Full-gate robustness requires $\mathbf{R} = 0$. The robust state-inversion pulse of Ref. [7], with $u_x = +\Omega$ for $3\pi/(2\Omega)$ followed by $u_x = -\Omega$ for $\pi/(2\Omega)$ and $u_y = 0$, has nominal gate $R_x(\pi)$ but $\mathbf{R} = -2\mathbf{e}_z$. It therefore retains a first-order full-gate error.

Every admissible full gate gives an absolutely continuous unit-vector curve with antipodal endpoints and speed at most one. Figures 1(a) and 1(b) show the two geometric views for a planar attaining example: $\mathbf{n}$ moves on the unit sphere, while its integral traces the error curve in Euclidean space. The first view constrains how quickly the detuning direction can turn; the second identifies the residual with the separation between the error-curve endpoints. The following theorem gives the smallest such residual under the full gate constraint.

Theorem 1 (Minimum detuning residual)**.** *For each fixed $L \in [\pi, 7\pi/3]$, among all controls satisfying (2) and the full nominal gate (3),*

$$\boxed{\min_u |\mathbf{R}| = g(L), \qquad g(L) = 4\cos\frac{L-\pi}{4} - 2.} \tag{9}$$

Apart from changes on sets of measure zero, all attaining controls have $u_y = 0$ and consist of the successive values $u_x = +\Omega, -\Omega, +\Omega$, or their overall negative, with durations

$$\frac{1}{\Omega}(a, 2a+\pi, a), \qquad a = \frac{L-\pi}{4}. \tag{10}$$

At $L = \pi$ the two outer segments have zero duration.

The minimum decreases from 2 at $L = \pi$ to zero at $L = 7\pi/3$. It therefore quantifies the best possible partial compensation at every duration before complete first-order cancellation becomes possible. We prove the geometric lower bound in Sec. III, then establish attainment and characterize all full-gate equality cases in Sec. IV.

III. PROOF OF THE GEOMETRIC LOWER BOUND

The key idea is tangent projection, as used in the proof of Schur's comparison theorem [23]. We choose the detuning direction at the midpoint of the gate as the projection axis. On each half of the trajectory, the speed limit bounds the angular distance from that midpoint direction. Integrating the resulting lower bound on the projection gives a lower bound on the residual norm. The argument uses only the antipodal endpoints and the speed limit, so it applies to a larger class of curves than those produced by the physical control model.

Theorem 2 (Antipodal spherical curves)**.** *Let $d \geq 2$ and $\pi \leq L \leq 7\pi/3$. For every absolutely continuous $\mathbf{n} : [0, L] \to S^{d-1}$ satisfying $|\mathbf{n}'| \leq 1$ almost everywhere and $\mathbf{n}(L) = -\mathbf{n}(0)$, set $\mathbf{v} = \mathbf{n}(L/2)$ and $\mathbf{R} = \int_0^L \mathbf{n}(\tau)\, \mathrm{d}\tau$. Then*

$$|\mathbf{R}| \geq \mathbf{v} \cdot \mathbf{R} \geq g(L) \geq 0. \tag{11}$$

The bound is attained for every L in the stated interval and every $d \geq 2$.

Proof. Set $c = L/2$ and let d_S denote spherical geodesic distance. For unit vectors $\mathbf{p}$ and $\mathbf{q}$, this distance is the angle between them:

$$d_S(\mathbf{p}, \mathbf{q}) = \arccos(\mathbf{p} \cdot \mathbf{q}).$$

The geodesic distance between two points is no greater than the length of any spherical path connecting them. The condition $|\mathbf{n}'| \leq 1$ bounds the length traveled during a time interval by the length of that interval. Consequently,

$$d_S(\mathbf{n}(\tau_1), \mathbf{n}(\tau_2)) \leq |\tau_2 - \tau_1|. \tag{12}$$

Let $\alpha = d_S(\mathbf{v}, \mathbf{n}(0))$. Because the endpoints are antipodal, their distances from $\mathbf{v}$ sum to π, so the other endpoint is at distance $\pi - \alpha$. Each endpoint must be reachable from the midpoint in time c. Thus both $\alpha \leq c$ and $\pi - \alpha \leq c$, which give

$$\max(0, \pi - c) \leq \alpha \leq \min(\pi, c). \tag{13}$$

To treat the two halves with the same argument, parameterize each from its outer endpoint toward the midpoint and call it $\mathbf{m} : [0, c] \to S^{d-1}$. Its initial distance from $\mathbf{v}$ is $\beta \in \{\alpha, \pi - \alpha\}$, and $\mathbf{m}(c) = \mathbf{v}$. At time t, the distance from $\mathbf{v}$ cannot exceed $\beta + t$, because the curve has traveled for only time t from its starting point. It also cannot exceed $c - t$, because that much time remains to reach $\mathbf{v}$. Finally, spherical geodesic distance never exceeds π. Combining the three restrictions gives the angular envelope

$$d_S(\mathbf{v}, \mathbf{m}(t)) \leq \min\{\beta + t, c - t, \pi\}. \tag{14}$$

The two affine bounds form a tent, which is truncated at height π if its peak exceeds that value; see Fig. 2. Since

the projection is $\mathbf{v}\cdot\mathbf{m}(t)=\cos d_S(\mathbf{v},\mathbf{m}(t))$ and cosine decreases on $[0,\pi]$, the largest allowed distance gives the smallest allowed projection. Integrating this lower bound yields

$$A(c,\beta):=\int_0^c \cos\big(\min\{\beta+t,c-t,\pi\}\big)\,\mathrm{d}t = \begin{cases} 2\sin\big((c+\beta)/2\big)-\sin\beta, & c+\beta\le 2\pi,\\ 2\pi-c-\beta-\sin\beta, & c+\beta\ge 2\pi. \end{cases} \tag{15}$$

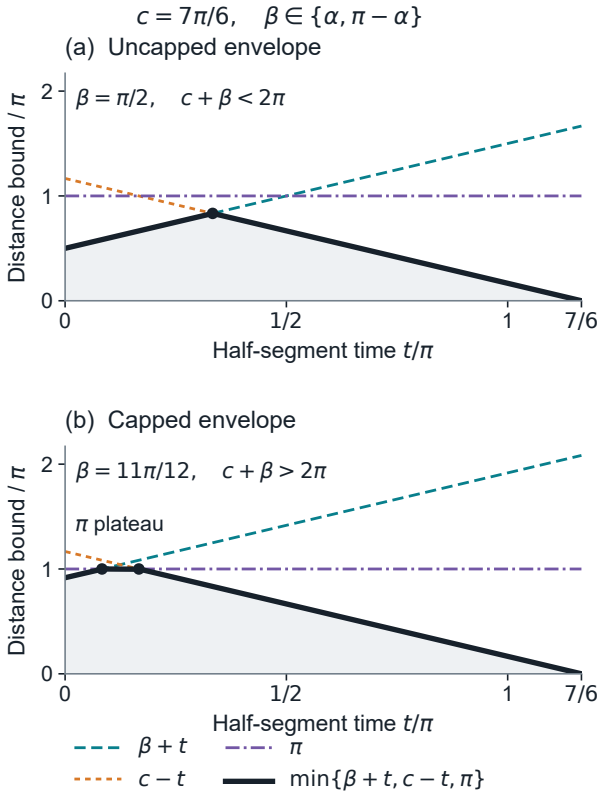


FIG. 2. Bounds on distance to the midpoint direction. Each half-curve is parameterized from its endpoint at $t=0$ toward the midpoint direction $\mathbf{v}$ at $t=c$. Its distance to $\mathbf{v}$ is at most $\min\{\beta+t,c-t,\pi\}$ (thick curve), where $\beta=\alpha$ for one half and $\beta=\pi-\alpha$ for the other. The three individual bounds are shown separately. For $c=7\pi/6$, (a) $\beta=\pi/2$ gives $c+\beta<2\pi$, so the two affine bounds meet below π; (b) $\beta=11\pi/12$ gives $c+\beta>2\pi$, so the spherical distance cap creates a plateau on $\pi-\beta\le t\le c-\pi$. Integrating the cosine of the envelope gives the two branches of $A(c,\beta)$.

The two branches have a direct geometric origin. The affine bounds meet at time $(c-\beta)/2$, which lies in the half-interval because $0\le\beta\le\min(\pi,c)$. Their common height is $(c+\beta)/2$. If this height is at most π, the envelope has only the two sloping pieces shown in Fig. 2(a). Otherwise a plateau of length $c+\beta-2\pi$ replaces the peak, as in Fig. 2(b). Integrating cosine over these pieces gives the two expressions in Eq. (15). Adding the half-integrals then gives

$$\mathbf{v}\cdot\mathbf{R}\ge P(c,\alpha):=A(c,\alpha)+A(c,\pi-\alpha). \tag{16}$$

It remains to minimize this scalar bound over the allowed position of the midpoint. First suppose neither half-envelope is truncated. Introduce

$$K=\sin\left(\frac{c}{2}+\frac{\pi}{4}\right),\qquad x=\cos\left(\frac{\alpha}{2}-\frac{\pi}{4}\right).$$

The stated ranges of c and α imply $1/2\le K\le 1$ and $1/\sqrt{2}\le x\le 1$. Using $\sin\alpha=2x^2-1$, the sum simplifies to

$$P(c,\alpha)=4Kx-4x^2+2. \tag{17}$$

Its difference from the desired lower bound factors as

$$P(c,\alpha)-(4K-2)=4(1-x)(1+x-K)\ge 0. \tag{18}$$

The second factor is strictly positive. Hence equality requires $x=1$, or $\alpha=\pi/2$: the midpoint direction must be equally distant from the two endpoints. For $c\le\pi$, neither half-envelope can reach above π, so this case already covers every feasible α.

For the remaining durations, one half-envelope may be truncated. Exchanging the endpoints leaves P unchanged, so take $\alpha\le\pi/2$. The half with initial distance α remains untruncated. The other half is truncated only when $c+(\pi-\alpha)\ge 2\pi$, leaving the region

$$\pi\le c\le 7\pi/6,\qquad 0\le\alpha\le c-\pi\le\pi/6. \tag{19}$$

Substituting the appropriate branch of Eq. (15) for each half gives

$$P(c,\alpha)=2\sin\frac{c+\alpha}{2}+\pi-c+\alpha-2\sin\alpha. \tag{20}$$

Because $(c+\alpha)/2\in[\pi/2,2\pi/3]$,

$$\partial_\alpha P=\cos\frac{c+\alpha}{2}+1-2\cos\alpha\le 1-\sqrt{3}<0. \tag{21}$$

The smallest value in this truncated region therefore occurs at its boundary, where the plateau has just disappeared:

$$P(c,\alpha)\ge P(c,c-\pi)>4K-2.$$

The last inequality follows from Eq. (18), which applies at the boundary. It is strict because $\alpha=c-\pi\le\pi/6$ cannot meet the equality condition $\alpha=\pi/2$. The truncated case thus cannot lower the bound or supply an additional equality case. Combining the two cases yields

$$P(c,\alpha)\ge 4\sin(c/2+\pi/4)-2=g(L), \tag{22}$$

with equality only at $\alpha=\pi/2$. Since $|\mathbf{v}|=1$ and $g(L)\ge 0$, this proves (11). The planar family constructed next attains the bound and embeds into every dimension $d\ge 2$. □

In the error-curve picture, the theorem bounds the endpoint displacement of every unit-speed curve of length L with curvature at most one and opposite endpoint tangents. A nonzero lower bound prevents that curve from closing. Its first zero, at $L = 7\pi/3$, is therefore the earliest duration at which the geometry permits first-order cancellation.

IV. ATTAINMENT AND ALL EQUALITY CASES

The geometric comparison gives a lower bound for every admissible control. To establish the minimum for a full quantum gate, we now construct controls that reach the bound and then show that every equality curve satisfying the complete propagator condition belongs to this family.

A. The attaining three-segment family

The attaining family uses the maximum drive amplitude along one fixed axis, reversing its sign twice. Its single parameter controls the duration and the degree of detuning compensation while preserving the nominal gate.

For $a = (L - \pi)/4 \in [0, \pi/3]$, take $u_y = 0$ and

$$\frac{u_x}{\Omega} = q_a(\tau) := \begin{cases} +1, & 0 \le \tau < a, \\ -1, & a \le \tau < 3a + \pi, \\ +1, & 3a + \pi \le \tau \le 4a + \pi. \end{cases} \tag{23}$$

The accumulated signed angle is

$$\phi(\tau) = \int_0^\tau q_a(\tau')\, \mathrm{d}\tau'.$$

Its successive segment-end values are $a, -a - \pi, -\pi$. The zero-detuning Hamiltonians commute, and hence

$$U_0(T) = R_x(-\pi) = -R_x(\pi). \tag{24}$$

The noise direction is

$$\mathbf{n}(\tau) = (0, \sin\phi(\tau), \cos\phi(\tau)). \tag{25}$$

Integration on the three segments gives

$$\begin{aligned} i \int_0^L e^{i\phi(\tau)}\, \mathrm{d}\tau &= 2e^{ia} - 2e^{-i(a+\pi)} + e^{-i\pi} - 1 \\ &= 4\cos a - 2 = g(L). \end{aligned} \tag{26}$$

The same family integral appears in Appendix D, Eq. (D2), of Ref. [10]. In dimensionless time, its cosine integral for target angle θ_f and positive switching angle Θ_{SB} reads

$$I_c := \int_0^L \cos\Theta(\tau)\, \mathrm{d}\tau = 4\sin\Theta_{\mathrm{SB}} - 2\sin\frac{\theta_f}{2}.$$

For the pulse used here, the shifted accumulated angle and parameters are

$$\Theta = -\phi - \frac{\pi}{2}, \qquad \theta_f = \pi, \qquad \Theta_{\mathrm{SB}} = \frac{\pi}{2} + a.$$

The angle Θ runs successively from $-\pi/2$ to $-\Theta_{\mathrm{SB}}$, then to $+\Theta_{\mathrm{SB}}$, and finally to $+\pi/2$. The integral is therefore $I_c = 4\cos a - 2$, consistent with Eq. (26).

Equation (26) gives $\mathbf{R} = -g(L)\mathbf{e}_y$. The midpoint satisfies $\phi(L/2) = -\pi/2$, so $\mathbf{v} = -\mathbf{e}_y$. Thus the residual is parallel to the projection direction, and the norm and projection bounds in Eq. (11) are both attained. Reversing the overall sign of the control preserves the residual norm and realizes the other sign of the nominal target.

At $a = \pi/3$, the family is short-CORPSE and the error curve closes, as shown in Fig. 3. At $a = 0$, it is a single rectangular π pulse. Thus the known robust construction and the shortest nominal gate are endpoints of one globally attaining family.

B. Geometric rigidity and uniqueness of the control

Equality in the integral bound is restrictive: each half of the direction curve must reach its distance envelope at every time. This condition fixes the curve to a single plane. The full gate condition then determines the orientation of that plane and hence the physical control.

Suppose $|\mathbf{R}| = g(L)$. Every step in

$$g(L) = |\mathbf{R}| \ge \mathbf{v}\cdot\mathbf{R} \ge P(c, \alpha) \ge g(L) \tag{27}$$

must be an equality, including at the zero $g(L) = 0$. Thus $\alpha = \pi/2$. The pointwise gaps in the two projected half-integrals are continuous and nonnegative, with total integral zero, so they vanish identically. Orient each half now from the midpoint towards its endpoint. Its distance from $\mathbf{v}$ must be

$$\theta(\ell) = \min\{\ell, c + \pi/2 - \ell\}, \qquad 0 \le \ell \le c. \tag{28}$$

For $\ell > 0$, write the half-curve as

$$\mathbf{m}(\ell) = \cos\theta(\ell)\, \mathbf{v} + \sin\theta(\ell)\, \mathbf{e}(\ell), \tag{29}$$

where $\mathbf{e} \perp \mathbf{v}$ and $|\mathbf{e}| = 1$. Since $0 < \theta(\ell) \le c/2 + \pi/4 \le 5\pi/6 < \pi$, the vector $\mathbf{e}$ is absolutely continuous on every $[\varepsilon, c]$. Almost everywhere on that interval,

$$|\mathbf{m}'|^2 = (\theta')^2 + \sin^2\theta\, |\mathbf{e}'|^2 \le 1. \tag{30}$$

The distance from $\mathbf{v}$ already changes at the full allowed rate, $|\theta'| = 1$ almost everywhere. Equation (30) therefore leaves no speed for motion in the orthogonal direction: $\mathbf{e}$ must be constant. As $\theta(c) = \pi/2$, the constants for the two halves are respectively $\mathbf{n}(0)$ and $-\mathbf{n}(0)$. Both halves therefore lie in the same great circle and have precisely the shape generated by (23). This also holds when $L = \pi$, where the outer segments disappear.

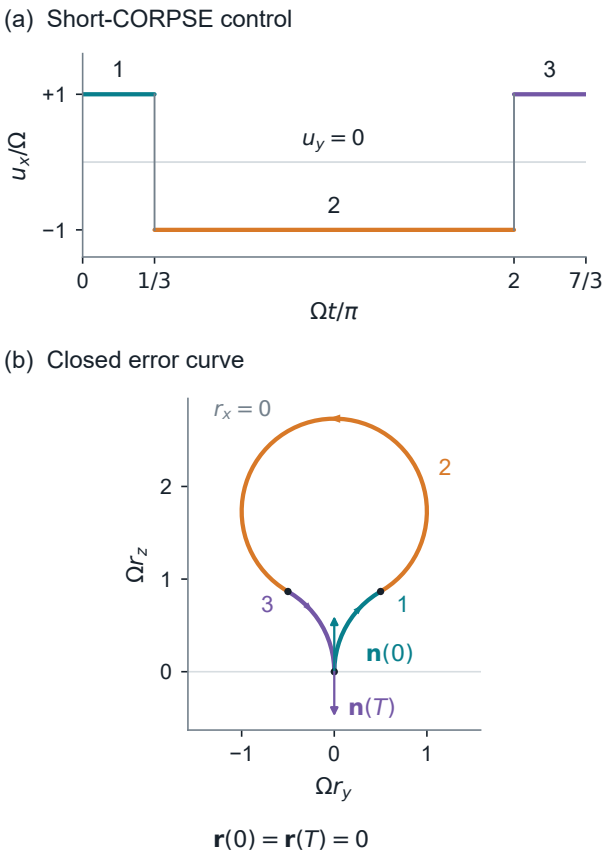


FIG. 3. Short-CORPSE control and error curve at the minimum first-order detuning-robust π-gate time $T = 7\pi/(3\Omega)$. (a) The standard short-CORPSE sequence [2] has saturated segments of durations $(\pi/3, 5\pi/3, \pi/3)/\Omega$, with $u_y = 0$. (b) The planar error curve consists of three unit-radius circular arcs in dimensionless coordinates $\Omega\mathbf{r}$. It returns to its starting point with opposite initial and final tangents, canceling the first-order detuning error. Numbers 1–3 and colors identify the same successive segments in both panels; arrows show the direction of traversal and dots mark the segment endpoints.

The full gate condition fixes the remaining plane orientation. Given $\mathbf{v} \perp \mathbf{e}_z$, set $\mathbf{a} = \mathbf{e}_z \times \mathbf{v}$. The fixed-axis evolution

$$U_a(t) = \exp\left[-\frac{i}{2}\phi(\Omega t)\mathbf{a}\cdot\boldsymbol{\sigma}\right]$$

generates the equality curve. To determine whether the same curve could come from a different control, suppose a transverse evolution U_0 generates the same $\mathbf{n}$. Then $W = U_0 U_a^\dagger$ commutes with σ_z and is diagonal in its eigenbasis. On the other hand,

$$\dot{W} = -iH_0 W + iWH_a, \tag{31}$$

where both nominal Hamiltonians are off-diagonal because no longitudinal control is available. Since W is diagonal, the right-hand side is off-diagonal, whereas $\dot{W}$ is diagonal. Consequently $\dot{W} = 0$ almost everywhere. With $W(0) = I$, this gives $W = I$. Finally, $U_a(T) = i\mathbf{a}\cdot\boldsymbol{\sigma}$ equals $\pm R_x(\pi)$ only for $\mathbf{a} = \pm\mathbf{e}_x$. Hence the controls in Theorem 1 exhaust every attaining case. Together with the midpoint bound and the explicit construction, this completes the proof of Theorem 1. The zero of the residual bound has a direct control interpretation: it is the earliest time at which the error curve can close, and the construction above realizes closure with the required full gate.

Corollary 1 (Minimum time for first-order cancellation)**.** *The minimum duration of an admissible control satisfying* (3) *and* $\partial_\delta U_\delta(T)|_{\delta=0} = 0$ *is*

$$T_* = \frac{7\pi}{3\Omega}. \tag{32}$$

Only short-CORPSE and its overall negative attain it, with segment durations $(\pi/3, 5\pi/3, \pi/3)/\Omega$.

Proof. Equation (5) makes first-order cancellation equivalent to $\mathbf{R} = 0$. The antipodal endpoint condition already requires $L \geq \pi$, and $g(L) > 0$ for $\pi \leq L < 7\pi/3$. Theorem 1 therefore excludes cancellation before $7\pi/(3\Omega)$ and shows that the two stated controls attain this time. Its equality classification also gives their uniqueness up to changes on sets of measure zero. □

V. PHYSICAL CONSEQUENCES OF THE RESIDUAL BOUND

The residual bound measures the propagator response to first order in detuning. To connect it to gate performance, we first derive the associated average gate infidelity. We then include weak Markovian dissipation, for which a longer gate incurs a larger leading loss. Balancing these contributions determines the degree of detuning compensation that minimizes the leading total gate error.

A. Detuning sensitivity and average gate infidelity

Consider first unitary evolution at a fixed static detuning. We use the standard average gate fidelity, obtained by averaging the state fidelity uniformly over all pure input states [24, 25]. For a fixed admissible control $u = (u_x, u_y)$, define the dimensionless detuning $f = \delta/\Omega$ and compare the actual gate with its nominal value:

$$\begin{aligned} V_f &= U_0(T)^\dagger U_{\Omega f}(T), \\ F_{\mathrm{avg}}[u; f] &= \frac{|\operatorname{Tr} V_f|^2 + 2}{6}, \\ \chi[u] &= \lim_{f\to 0} \frac{1 - F_{\mathrm{avg}}[u; f]}{f^2}. \end{aligned} \tag{33}$$

The coefficient χ is the leading infidelity per squared dimensionless detuning. It is related to the gate fidelity

susceptibility of Ref. [3] by a change of fidelity and detuning conventions. Explicitly, if

$$F_U = \frac{|\operatorname{Tr} V_f|^2}{4}, \qquad 1 - F_U = \chi_U \delta^2 + o(\delta^2),$$

then

$$F_{\text{avg}} = \frac{2F_U + 1}{3}, \qquad \chi = \frac{2\Omega^2}{3}\chi_U.$$

To obtain χ from the error curve, write the relative propagator in its $SU(2)$ form,

$$U_0(T)^\dagger U_\delta(T) = w_\delta I - i\mathbf{b}_\delta \cdot \boldsymbol{\sigma},$$
$$w_\delta^2 + |\mathbf{b}_\delta|^2 = 1,$$

where w_δ and $\mathbf{b}_\delta$ are real. Equation (5) fixes its vector part to first order in detuning:

$$\mathbf{b}_\delta = \frac{\delta}{2}\mathbf{r}(T) + O(\delta^2).$$

Substituting into the fidelity gives

$$1 - F_{\text{avg}} = \frac{2}{3}|\mathbf{b}_\delta|^2 = \frac{\delta^2}{6}|\mathbf{r}(T)|^2 + O(\delta^3),$$
$$\chi[u] = \frac{\Omega^2}{6}|\mathbf{r}(T)|^2. \tag{34}$$

Thus the propagator response to first order in detuning produces a quadratic contribution to the infidelity, with coefficient proportional to the squared endpoint displacement. In particular, $\chi = 0$ is equivalent to the vanishing of the term linear in detuning in the full propagator and to closure of the error curve.

Theorem 1 now determines the minimum sensitivity at each duration:

$$\boxed{\chi_{\min}(L) = \frac{g(L)^2}{6} = \frac{1}{6}\left[4\cos\frac{L-\pi}{4} - 2\right]^2}. \tag{35}$$

The controls in Eq. (23) and their overall negative attain this speed–sensitivity frontier throughout $\pi \le L \le 7\pi/3$. The frontier decreases from $\chi_{\min}(\pi) = 2/3$ for the rectangular π pulse to zero for short-CORPSE. It gives a quantitative limit on the fidelity improvement available from extra control time; access to a second transverse axis cannot improve that limit. As an intermediate example,

$$\chi_{\min}(2\pi) = \frac{(2\sqrt{2}-2)^2}{6} \simeq 0.11438.$$

This gate uses 6/7 of the time required for complete first-order cancellation while substantially reducing the sensitivity relative to a rectangular pulse.

B. Weak dissipation and the joint error budget

To include the cost of spending more time under control, assume Markovian relaxation and pure dephasing with rates that are independent of the drive. The static detuning is drawn from a symmetric, zero-mean quasistatic distribution and is independent of these dissipative processes. The dynamics take the standard Lindblad form [26, 27],

$$\dot\rho = -i[H_\delta, \rho] + \Gamma_1 \mathcal{D}[\sigma_-]\rho + \frac{\Gamma_\phi}{2}\mathcal{D}[\sigma_z]\rho, \tag{36}$$

with

$$\mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\},$$
$$\sigma_- = \frac{\sigma_x - i\sigma_y}{2}.$$

In this convention, Γ_ϕ is the pure-dephasing rate and the transverse decay rate is $\Gamma_2 = \Gamma_1/2 + \Gamma_\phi$. The average fidelity now compares the noisy quantum channel with the same nominal unitary gate.

First set $\delta = 0$ and retain the first order in the decay rates. The homogeneous damping matrix in the Bloch equation is

$$D = \operatorname{diag}(-\Gamma_2, -\Gamma_2, -\Gamma_1).$$

Let $O_0(t)$ be the ideal Bloch rotation and $\Gamma = \Gamma_1 + \Gamma_\phi$. In the ideal rotating frame, the homogeneous part of the channel is

$$M = I + \int_0^T O_0(t)^{\mathsf{T}} D O_0(t)\,\mathrm{d}t + O((\Gamma T)^2).$$

For an affine Bloch map, the average fidelity relative to the identity is [24, 25]

$$\mathbf{q} \mapsto M\mathbf{q} + \mathbf{c}, \qquad F_{\text{avg}} = \frac{1}{2} + \frac{\operatorname{Tr} M}{6}.$$

The translation $\mathbf{c}$ makes no contribution because the average input Bloch vector vanishes. Since rotations preserve $\operatorname{Tr} D$, the resulting first-order loss is

$$1 - F_{\text{avg}} = \frac{\Gamma_1 + \Gamma_\phi}{3}T + O((\Gamma T)^2), \qquad \Gamma = \Gamma_1 + \Gamma_\phi. \tag{37}$$

This is the single-qubit specialization of the universal first-order fidelity reduction [22]. For this noise model, changing the waveform rotates the damping directions but does not change their trace. The leading dissipative loss therefore depends only on the gate duration. This is the physical cost that competes with detuning compensation.

To combine the two errors at the same perturbative order, take the weak-noise limit

$$\delta = \eta\xi, \qquad \Gamma_j = \eta^2\gamma_j \quad (j = 1, \phi), \qquad \eta \to 0.$$

Here the γ_j, the control, and its bounded duration are fixed, and the symmetric distribution of ξ has a finite fourth moment. This scaling makes the quadratic detuning loss and the linear dissipative loss both of order η^2. Define their dimensionless strengths by

$$s^2 = \frac{\operatorname{Var}(\delta)}{\Omega^2}, \qquad \kappa = \frac{\Gamma_1 + \Gamma_\phi}{3\Omega}. \tag{38}$$

In the ensemble average, symmetry removes odd powers of δ, including the cubic detuning term and the leading mixed detuning–dissipation term. The first omitted contributions are of order η^4: fourth order in detuning, second order in the decay rates, or second order in detuning together with first order in a decay rate. The finite fourth moment controls the averaged detuning remainder. Combining the static-error expansion with Eq. (37) therefore gives

$$\mathbb{E}[1 - F_{\mathrm{avg}}] = s^2\chi[u] + \kappa L + O(\eta^4). \tag{39}$$

For the attaining single-axis family, the unitary fidelity is already even in δ before averaging. Indeed,

$$H_{-\delta} = \sigma_x H_\delta \sigma_x,$$

and the nominal target commutes with σ_x.

Equation (39) separates the two leading physical losses: $s^2\chi[u]$ measures sensitivity to the spread of static detunings, while κL measures exposure to dissipation. The $O(\eta^4)$ remainder contains the higher-order effects excluded from this budget; in particular, zero first-order sensitivity need not mean zero infidelity at finite noise. Leading-order addition of coherent and dissipative losses also appears in weak-dissipation expansions [22] and comparisons of short and robust Rydberg gates [28].

At a fixed $L \in [\pi, 7\pi/3]$, all admissible waveforms have the same leading dissipative cost. Minimizing the total leading loss therefore requires precisely the minimum sensitivity already found:

$$\boxed{\mathcal{E}_{\min}(L) = s^2\chi_{\min}(L) + \kappa L.} \tag{40}$$

The three-segment family (23) attains this value. The remaining choice is how long to run that pulse, which determines the balance between the two error contributions.

C. Balancing detuning suppression and dissipation

Canceling the first-order detuning response removes the quadratic loss, but it requires the longest gate on the frontier. When dissipation is present, the relevant objective is the sum of the remaining detuning loss and the dissipative loss accumulated during the pulse. We therefore minimize Eq. (40) over duration. The frontier is attainable and globally optimal at each duration, so this one-variable minimization covers every admissible waveform.

For $\kappa > 0$, it is enough to consider $\pi \le L \le 7\pi/3$. The lower endpoint is fixed by the nominal gate, and any longer duration than $7\pi/3$ has a dissipative cost alone greater than $\kappa(7\pi/3)$, the complete leading budget at the robust endpoint. Thus a pulse longer than short-CORPSE cannot improve this objective, regardless of its detuning sensitivity.

For $s^2 > 0$, introduce the ratio of dissipative to detuning cost,

$$\lambda = \frac{\kappa}{s^2}, \qquad J(L) = \frac{\mathcal{E}_{\min}(L)}{s^2} = \chi_{\min}(L) + \lambda L.$$

The noise strengths fix λ, and the duration selects a point on this curve. At an interior optimum, the reduction in detuning sensitivity obtained from a little more time must balance the additional dissipative cost:

$$J'(L) = \lambda - \frac{g(L)}{3}\sin\frac{L-\pi}{4} = 0. \tag{41}$$

The global choice can be characterized explicitly. Write $a = (L-\pi)/4 \in [0, \pi/3]$ and set

$$p(a) = \frac{2}{3}(2\cos a - 1)\sin a. \tag{42}$$

Since $p'(a) = \frac{2}{3}(4\cos^2 a - \cos a - 2)$, p increases up to $a_m = \arccos[(1+\sqrt{33})/8]$ and then decreases to zero. For $\lambda > 0$, the rectangular endpoint is a local minimum, the robust endpoint is not a minimum, and only the larger root of $p(a) = \lambda$ can be an interior minimum. At such a stationary point, the budget excess over the rectangular pulse is

$$\Delta(a) = \chi_{\min}(\pi + 4a) + 4ap(a) - \frac{2}{3}.$$

The identity $\Delta'(a) = 4ap'(a)$, together with $\Delta(0) = 0$ and $\Delta(\pi/3) = -2/3$, shows that Δ has exactly one zero a_c on $(a_m, \pi/3)$. Equivalently,

$$a_c(2\cos a_c - 1)\sin a_c = \cos a_c(1 - \cos a_c). \tag{43}$$

The associated threshold and duration are

$$\begin{aligned} \lambda_c &= p(a_c) \simeq 0.17625857, \\ L_c &= \pi + 4a_c \simeq 2.04725676\pi. \end{aligned} \tag{44}$$

For $0 < \lambda < \lambda_c$, the unique optimal duration is the interior minimum. At $\lambda = \lambda_c$, $L = \pi$ and $L = L_c$ are both global minima; for $\lambda > \lambda_c$, the unique optimal duration is $L = \pi$. The optimal duration thus jumps to π at this threshold, even though all compensation levels in the pulse family are available.

Figure 4 shows the sensitivity frontier and the gain from selecting an intermediate compensation level. To compare with the better of the two endpoint pulses, define, for $\lambda > 0$,

$$\begin{aligned} B(\lambda) &= \min\left\{\frac{2}{3} + \pi\lambda, \frac{7\pi}{3}\lambda\right\}, \\ G(\lambda) &= 1 - \frac{\min_{\pi \le L \le 7\pi/3} J(L)}{B(\lambda)}. \end{aligned} \tag{45}$$

The endpoint budgets cross at $\lambda_0 = 1/(2\pi)$. For each fixed L, the ratio $J(L)/B(\lambda)$ is nonincreasing before this crossing. After it, the derivative has the sign of $(2/3)L -$

$\pi\chi_{\min}(L) \geq 0$, since $L \geq \pi$ and $0 \leq \chi_{\min} \leq 2/3$. Taking the minimum over L therefore places the maximum gain at λ_0. The gain there is positive: the endpoint budgets are equal, and $J'(7\pi/3) = \lambda_0 > 0$ makes a slightly shorter pulse better. At any other ratio with positive gain, the optimizing duration is interior. Evaluating that same pulse at λ_0 strictly lowers its budget ratio. The minimum ratio at λ_0 is no larger than this improved value, proving that the positive maximum is unique. Numerically,

$$\begin{aligned} G(\lambda_0) &\simeq 4.91\%, \\ L_{\mathrm{opt}}(\lambda_0) &\simeq 2.08\pi. \end{aligned} \tag{46}$$

At this ratio, both endpoint budgets equal $7/6$. The gain comes from retaining most of the detuning suppression while reducing exposure to dissipation.

The local behavior near the robust endpoint explains why some shortening is always beneficial when $\kappa > 0$. Write $L_* = 7\pi/3$ and let $h = L_* - L \downarrow 0$. The sensitivity grows quadratically with the reduction in duration:

$$\chi_{\min}(L_* - h) = \frac{h^2}{8} + O(h^3), \tag{47}$$

and therefore

$$\mathcal{E}_{\min}(L_* - h) - \mathcal{E}_{\min}(L_*) = -\kappa h + \frac{s^2 h^2}{8} + O(s^2 h^3). \tag{48}$$

The saved dissipative loss is linear in h, so it exceeds the quadratic detuning penalty for sufficiently small shortening. Consequently, full first-order cancellation does not minimize this budget at any positive dissipation rate. When dissipation is weak relative to detuning, the best duration remains close to the robust endpoint:

$$h = 4\lambda + O(\lambda^2), \qquad \lambda \to 0^+.$$

The relation to finite-noise gate fidelity follows from the orders in Eq. (39). Both leading losses are of order η^2, whereas the omitted contribution is of order η^4. For two fixed pulse choices at fixed relative noise strengths, a nonzero difference in the leading budget therefore determines their fidelity ordering as the noise weakens. If their leading budgets coincide, higher-order terms are needed to distinguish them. The scalar optimization identifies the best leading-order operating point and makes explicit the level of the weak-noise expansion at which that choice is resolved.

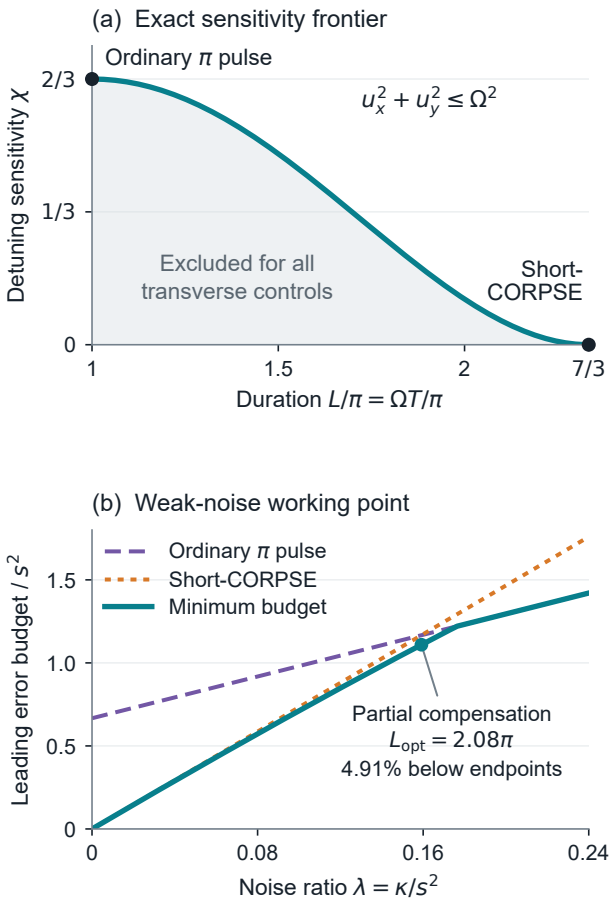


FIG. 4. Exact speed–sensitivity frontier and its weak-noise application. (a) The minimum coefficient $\chi = \lim_{f\to 0}(1 - F_{\mathrm{avg}})/f^2$, with $f = \delta/\Omega$, for a full nominal π gate at dimensionless duration $L = \Omega T$. The shaded region is excluded for every measurable transverse control satisfying $u_x^2 + u_y^2 \leq \Omega^2$. The boundary is attained by the same-axis three-segment family [10], joining the ordinary π pulse at $L = \pi$ to short-CORPSE [2] at $L = 7\pi/3$. (b) Leading error budget divided by s^2 for zero-mean quasistatic detuning and drive-independent weak Markovian relaxation and pure dephasing: $\chi_{\min}(L) + \lambda L$, where $s^2 = \mathrm{Var}(\delta)/\Omega^2$, $\lambda = \kappa/s^2$, and $\kappa = (\Gamma_1 + \Gamma_\phi)/(3\Omega)$. The solid curve minimizes this budget over duration; the dashed and dotted curves use the two endpoint pulses. At $\lambda = 1/(2\pi)$, the endpoints have equal budgets, while partial compensation with $L_{\mathrm{opt}} \simeq 2.08\pi$ lowers the leading budget by 4.91%, the maximum relative reduction over $\lambda > 0$ compared with the better endpoint pulse at this order. The optimization concerns the weak-noise coefficient.

VI. DISCUSSION

The duration-dependent bound makes the time cost of detuning compensation quantitative. Its first zero fixes the factor $7/3$ between the shortest first-order robust gate and the shortest nominal gate. The positive part supplies information that a minimum robust time alone cannot provide: it determines the residual that must remain when

the allowed duration is shorter. Through Eq. (34), the same curve gives the corresponding unavoidable quadratic infidelity. The known three-segment family therefore provides both a practical pulse at each duration and a benchmark that no other admissible two-axis control can improve.

The mathematical mechanism is the common midpoint constraint on the two halves of the direction curve. A bounded drive prevents either half from moving arbitrarily far from the midpoint direction and still reaching its prescribed endpoint. The resulting projected integral stays positive until the robust time. This comparison acts directly on every admissible curve, avoiding a classification of extremal trajectories or an assumption that an optimal control exists. It applies to arbitrary measurable controls. We prove that the attaining three-segment pulse family gives the global minimum over all admissible two-axis controls and exhausts every equality case.

The equality conditions explain the simple form of the optimal control. To reach the lower projection bound, the angular distance must change at its maximum allowed rate. No remaining speed is available for motion out of the plane, which forces the spherical curve onto a great circle. The full unitary target then selects its axis. Thus the lack of improvement from a second transverse control, reduced amplitudes, or additional switches follows from the equality structure of the bound. The complete gate condition is essential at this last step: opposite endpoint noise directions alone do not fix the action on every input state. For other target angles, both the endpoint geometry and the full-frame constraint change and require a separate analysis.

The physical role of the continuous frontier is clearest when two error mechanisms favor different durations. Static detuning favors more compensation, whereas drive-independent weak Markovian dissipation penalizes the extra time. Because the latter loss is independent of the waveform to leading order, the geometric result reduces their competition to the analytic scalar budget in Eq. (40). The quadratic onset of sensitivity near the robust endpoint means that a small reduction in compensation can save more dissipative error than it adds in detuning error. Partial compensation is then a predictable choice based on the relative noise strengths, using the same pulse family as complete cancellation.

The attaining pulses share one control axis; at zero detuning their net signed rotation is $-\pi$ (or $+\pi$ for the overall negative), so they do not cancel a common multiplicative amplitude error. The nontrivial three-segment pulses also require instantaneous phase reversals. A finite slew-rate constraint reduces the admissible control class, preserving the lower bound while potentially preventing its attainment. If dissipation depends on the drive, its cost can vary with the waveform even at fixed duration; a pulse that minimizes the detuning residual need then no longer minimize the total error. For noise outside the quasistatic regime, the error model must also be adjusted.

VII. CONCLUSION

For a full nominal qubit π gate under a joint transverse amplitude bound, we have determined the smallest first-order detuning residual at every duration from π/Ω to $7\pi/(3\Omega)$. With $L = \Omega T \in [\pi, 7\pi/3]$, midpoint projection gives the bound

$$|\mathbf{R}| \geq g(L) = 4\cos\frac{L-\pi}{4} - 2$$

for every admissible measurable control. Only the stated single-axis three-segment family and its overall negative attain this bound. Its first zero establishes $7\pi/(3\Omega)$ as the minimum time for first-order detuning cancellation. Using the relation between the residual and the quadratic coefficient of average gate infidelity in Eq. (34), we obtain the global minimum of this coefficient at fixed duration:

$$\begin{aligned}\chi_{\min}(L) = \min_u \chi[u] &= \frac{g(L)^2}{6} \\ &= \frac{1}{6}\left[4\cos\frac{L-\pi}{4} - 2\right]^2.\end{aligned}$$

With weak, drive-independent Markovian dissipation, the dimensionless duration and three-segment compensation parameter that minimize the leading total gate error satisfy

$$\begin{aligned}L_{\text{opt}} &\in \operatorname*{arg\,min}_{\pi\leq L\leq 7\pi/3} [s^2\chi_{\min}(L) + \kappa L], \\ a_{\text{opt}} &= \frac{L_{\text{opt}} - \pi}{4}.\end{aligned}$$

Here s^2 and κ are the noise parameters defined in Eq. (38), and a_{opt} fixes the three segment durations through Eq. (10). For $\kappa > 0$, the linear saving in dissipative loss from slightly shortening a fully compensated gate outweighs the quadratic detuning penalty. For $s^2 > 0$, as the relative cost $\lambda = \kappa/s^2$ increases past λ_c, the optimal dimensionless duration jumps to the rectangular-pulse value $L = \pi$.

The main mathematical results of this work have been formally proved in Lean.

ACKNOWLEDGMENTS

We used OpenAI Codex (GPT-6 Astra) to assist in translating the mathematical proofs in this paper into Lean code and to check the manuscript for grammatical errors. All AI outputs were reviewed by the authors.

DATA AND CODE AVAILABILITY

The Lean code, numerical scripts, figure data, and reproduction instructions are archived on Zenodo at DOI: 10.5281/zenodo.24154006.

[1] H. K. Cummins and J. A. Jones, Use of composite rotations to correct systematic errors in NMR quantum computation, New J. Phys. **2**, 6 (2000).

[2] H. K. Cummins, G. Llewellyn, and J. A. Jones, Tackling systematic errors in quantum logic gates with composite rotations, Phys. Rev. A **67**, 042308 (2003).

[3] P. M. Poggi, G. De Chiara, S. Campbell, and A. Kiely, Universally Robust Quantum Control, Phys. Rev. Lett. **132**, 193801 (2024).

[4] L. Balada Gaggioli, Time-optimal robust quantum control for partially known errors (2026), arXiv:2610.21596 [quant-ph].

[5] X. Cao, J. Cui, M. H. Yung, and R.-B. Wu, Robust Control of Single-Qubit Gates at the Quantum Speed Limit, Phys. Rev. A **110**, 022603 (2024).

[6] E. Bataille, G. Nomura, and M. Endres, Detuning- and Stark-robust Rydberg gates (2026), arXiv:2608.27426 [quant-ph].

[7] L. Van Damme, Q. Ansel, S. J. Glaser, and D. Sugny, Robust optimal control of two-level quantum systems, Phys. Rev. A **95**, 063403 (2017).

[8] S. Kukita, H. Kiya, and Y. Kondo, General off-resonance-error-robust symmetric composite pulses with three elementary operations, Phys. Rev. A **106**, 042613 (2022).

[9] Q. Ansel, S. J. Glaser, and D. Sugny, Selective and robust time-optimal rotations of spin systems, J. Phys. A: Math. Theor. **54**, 085204 (2021).

[10] S. Kukita, H. Kiya, and Y. Kondo, Optimal quantum controls robust against detuning error, Phys. Scr. **99**, 115124 (2024).

[11] J. Zeng and E. Barnes, Fastest pulses that implement dynamically corrected single-qubit phase gates, Phys. Rev. A **98**, 012301 (2018).

[12] J. Zeng, C. H. Yang, A. S. Dzurak, and E. Barnes, Geometric formalism for constructing arbitrary single-qubit dynamically corrected gates, Phys. Rev. A **99**, 052321 (2019).

[13] E. Piliouras, D. Lucarelli, and E. Barnes, An automated geometric space curve approach for designing dynamically corrected gates, npj Quantum Inf. **12**, 46 (2026).

[14] M. Sigalotti and Y. Chitour, Dubins' Problem on Surfaces II: Nonpositive Curvature, SIAM J. Control Optim. **45**, 457 (2006).

[15] H.-S. Kim and O. Cheong, The cost of bounded curvature, Comput. Geom. **46**, 648 (2013).

[16] E. J. Cockayne and G. W. C. Hall, Plane Motion of a Particle Subject to Curvature Constraints, SIAM J. Control **13**, 197 (1975).

[17] V. S. Patsko, S. G. Pyatko, and A. A. Fedotov, Three-dimensional reachability set for a nonlinear control system, J. Comput. Syst. Sci. Int. **42**, 320 (2003).

[18] H. J. Sussmann, Shortest 3-dimensional paths with a prescribed curvature bound, in *Proceedings of the 34th IEEE Conference on Decision and Control*, Vol. 4 (1995) pp. 3306–3312.

[19] J. Bae, J. H. Bai, B.-Y. Lee, J.-Y. Lee, and C.-H. Lee, On the Reachability of 3-Dimensional Paths with a Prescribed Curvature Bound, Automatica **178**, 112359 (2025).

[20] J. Bae and H.-L. Choi, Fixed-Time State Transfer via Pontryagin Extremals, Automatica **192**, 113146 (2026).

[21] R. Moser, Structure and classification results for the ∞-elastica problem, Amer. J. Math. **144**, 1299 (2022).

[22] T. Abad, J. Fernández-Pendás, A. Frisk Kockum, and G. Johansson, Universal Fidelity Reduction of Quantum Operations from Weak Dissipation, Phys. Rev. Lett. **129**, 150504 (2022).

[23] J. M. Sullivan, Curves of Finite Total Curvature, in *Discrete Differential Geometry*, Oberwolfach Seminars, Vol. 38, edited by A. I. Bobenko, P. Schröder, J. M. Sullivan, and G. M. Ziegler (Birkhäuser, 2008) pp. 137–161.

[24] M. D. Bowdrey, D. K. L. Oi, A. J. Short, K. Banaszek, and J. A. Jones, Fidelity of single qubit maps, Phys. Lett. A **294**, 258 (2002).

[25] M. A. Nielsen, A simple formula for the average gate fidelity of a quantum dynamical operation, Phys. Lett. A **303**, 249 (2002).

[26] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, Completely positive dynamical semigroups of N-level systems, J. Math. Phys. **17**, 821 (1976).

[27] G. Lindblad, On the generators of quantum dynamical semigroups, Commun. Math. Phys. **48**, 119 (1976).

[28] C. Fromonteil, D. Bluvstein, and H. Pichler, Protocols for Rydberg Entangling Gates Featuring Robustness against Quasistatic Errors, PRX Quantum **4**, 020335 (2023).